

COURSE CODE/TITLE:	CPE212 - Automating Server Management	SECTION:	CPE31S2
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ACTIVITY NO. 10	Install, Configure, and Manage Log Monitoring tool
1. PROCEDURE OUTPUT	

1. Create a playbook that:
 - a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)
2. Apply the concept of creating roles.
3. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)

Step 0: Create a directory for the HOA within the local repository.

```

user123@workstation:~$ cd CPE212_fegalan
user123@workstation:~/CPE212_fegalan$ ls
HOA4 HOA5 HOA6 HOA7 HOA8 notes.txt QUIZ7.1 README.md
user123@workstation:~/CPE212_fegalan$ mkdir HOA10
user123@workstation:~/CPE212_fegalan$ ls
HOA10 HOA4 HOA5 HOA6 HOA7 HOA8 notes.txt QUIZ7.1 README.md
user123@workstation:~/CPE212_fegalan$ cd HOA10

```

Step 1: Create inventory

Do nano inventory.yaml on the main directory aka HOA10

```

GNU nano 7.2
all:
  children:
    ubuntu_server:
      hosts:
        server1:
          ansible_host: 192.168.0.102

    centos_server:
      hosts:
        server2:
          ansible_host: 192.168.0.103
          ansible_user: centos_chris

    elasticsearch:
      children:
        ubuntu_server:
        centos_server:

    logstash:
      children:
        ubuntu_server:
        centos_server:

    kibana:
      children:
        ubuntu_server:
        centos_server:

```

Step 2: Create roles directory

```
user123@workstation:~/CPE212_fegalan/H0A10$ mkdir -p roles/elasticsearch/tasks
user123@workstation:~/CPE212_fegalan/H0A10$ mkdir -p roles/logstash/tasks
user123@workstation:~/CPE212_fegalan/H0A10$ mkdir -p roles/kibana/tasks
user123@workstation:~/CPE212_fegalan/H0A10$ ls
roles
```

- Set up elasticsearch, logstash, and kibana roles for later use.

Step 3: Create the main playbook for running the log monitoring tool.

Do nano logmonitoring.yaml in the main directory

```
GNU nano 7.2 logmonitoring.yaml
--
- hosts: all
  become: true
  tasks:
    - name: Create apt keyring directory
      ansible.builtin.file:
        path: /etc/apt/keyrings
        state: directory
        mode: "0755"
      when: ansible_os_family == "Debian"

    - name: Download Elastic GPG key
      ansible.builtin.get_url:
        url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
        dest: /etc/apt/keyrings/elastic.asc
        mode: "0644"
      when: ansible_os_family == "Debian"

    - name: Add Elastic apt repository
      ansible.builtin.apt_repository:
        repo: "deb [signed-by=/etc/apt/keyrings/elastic.asc] https://artifacts.elastic.co/packages/8.x/apt stable main"
        filename: elastic-8.x
        state: present
      when: ansible_os_family == "Debian"

    - name: Import Elastic GPG key
      ansible.builtin.rpm_key:
        key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
        state: present
      when: ansible_os_family == "RedHat"

    - name: Add Elastic yum repository
      ansible.builtin.yum_repository:
        name: elastic-8.x
        description: Elastic repository for 8.x packages
        baseurl: https://artifacts.elastic.co/packages/8.x/yum
        gpgcheck: true
        gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
        enabled: true
      when: ansible_os_family == "RedHat"

- hosts: elasticsearch
  become: true
  roles:
    - elasticsearch

- hosts: logstash
  become: true
  roles:
    - logstash

- hosts: kibana
  become: true
  roles:
    - kibana
```

- This playbook first adds the Elastic package repository and its security key to every server, using apt on Ubuntu and yum on CentOS, with the when conditions making each task run only on the matching OS. It then runs three plays that apply the elasticsearch, logstash and kibana roles to their own inventory groups. Which are all needed since they install and start the tools that store, process and display the logs.

Step 4: Creating tasks for elasticsearch

Inside /roles/elasticsearch/tasks/

Do nano main.yaml and type this out:

```
- name: Install Elasticsearch
  ansible.builtin.package:
    name: elasticsearch
    state: present

- name: Configure Elasticsearch
  ansible.builtin.copy:
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: root
    group: elasticsearch
    mode: "0660"
    content: |
      path.data: /var/lib/elasticsearch
      path.logs: /var/log/elasticsearch
      network.host: 0.0.0.0
      discovery.type: single-node
      xpack.security.enabled: false

- name: Elasticsearch User config permissions
  ansible.builtin.file:
    path: /etc/elasticsearch
    owner: elasticsearch
    group: elasticsearch
    recurse: true

- name: Enable and start Elasticsearch
  ansible.builtin.service:
    name: elasticsearch
    state: started
    enabled: true
```

Step 5: Creating tasks for logstash

Inside `/roles/logstash/tasks/`

Do nano `main.yaml` and type this out:

```
GNU nano 7.2 main.yaml
---
- name: Install Logstash
  ansible.builtin.package:
    name: logstash
    state: present

- name: Configure Logstash pipeline
  ansible.builtin.copy:
    dest: /etc/logstash/conf.d/logstash.conf
    mode: "0644"
    content: |
      input {
        beats {
          port => 5044
        }
      }
      output {
        elasticsearch {
          hosts => ["http://127.0.0.1:9200"]
          index => "logstash-%{+YYYY.MM.dd}"
        }
      }

- name: Enable and start Logstash
  ansible.builtin.service:
    name: logstash
    state: restarted
    enabled: true
```

- When creating this file the hosts may vary from user to user.

Step 6: Creating tasks for kibana

Inside `/roles/kibana/tasks/`

Do nano `main.yaml` and type this out:

```
GNU nano 7.2
---
- name: Install Kibana
  ansible.builtin.package:
    name: kibana
    state: present

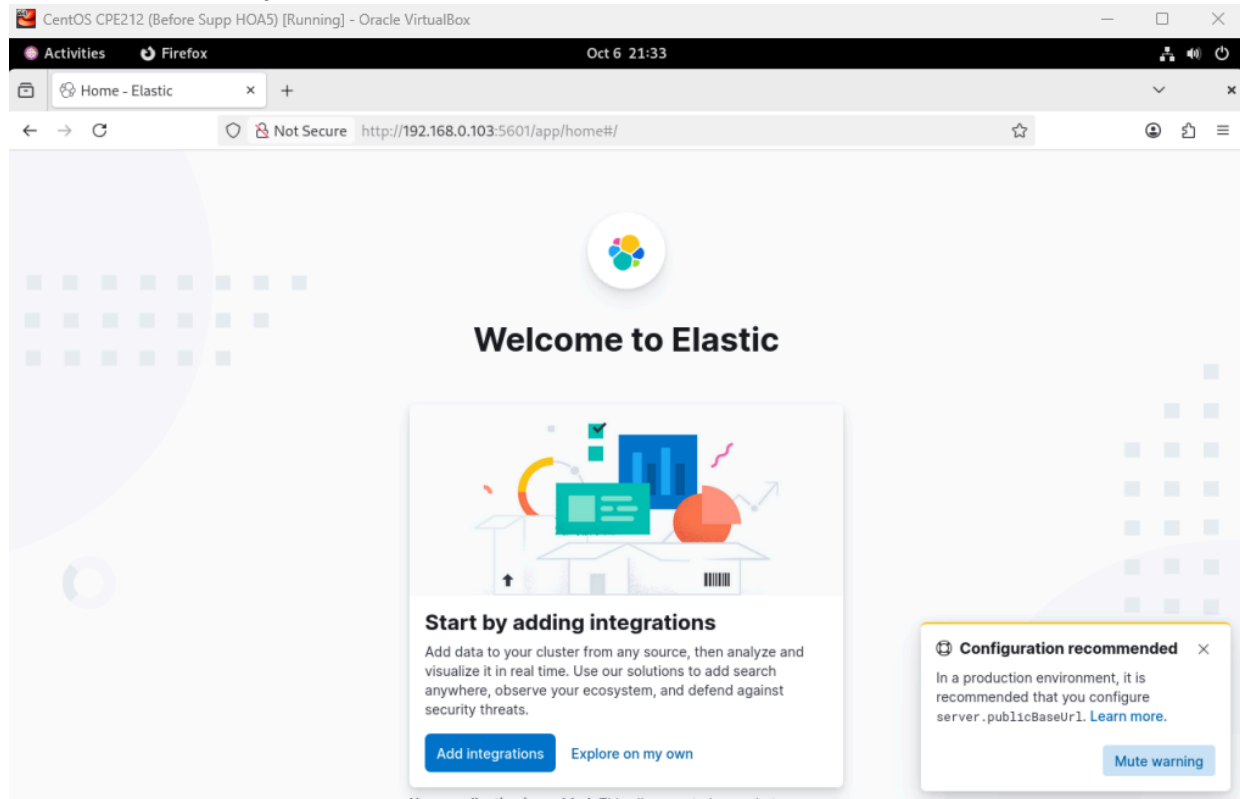
- name: Configure Kibana
  ansible.builtin.copy:
    dest: /etc/kibana/kibana.yml
    owner: root
    group: kibana
    mode: "0660"
    content: |
      server.host: "0.0.0.0"
      server.port: 5601
      elasticsearch.hosts: ["http://127.0.0.1:9200"]

- name: Enable and start Kibana
  ansible.builtin.service:
    name: kibana
    state: restarted
    enabled: true
```

4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.

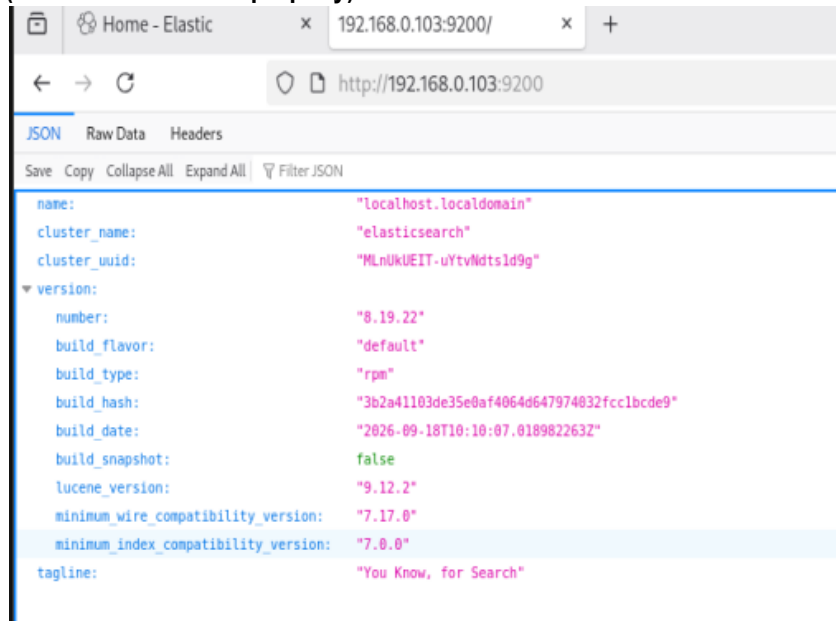
CENTOS:

(Kibana Works Properly)



Open the firefox browser and type your ip and :5601 port

(Elasticsearch works properly)



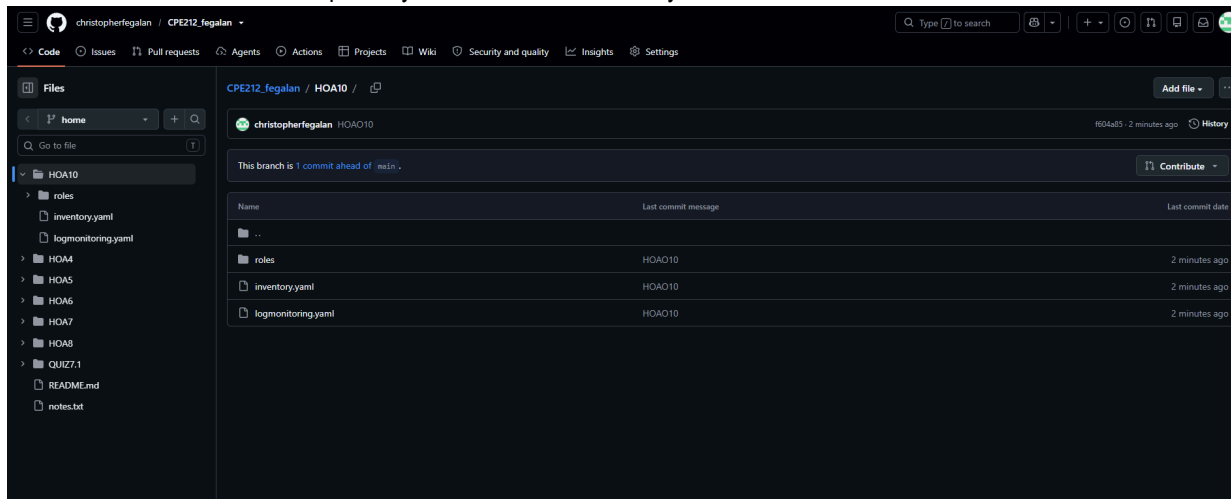
(The stack working properly)

```
active
[centos_chris@localhost ~]$ systemctl is-active elasticsearch kibana logstash
active
active
active
```

UBUNTU:

(playbook works completely fine) - The issue i couldn't get it to work is because it's so resource intensive my ubuntu virtual machine wont run anymore.

5. Make sure to create a new repository in GitHub for this activity.



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. What are the benefits of having log monitoring tool

- Having a log monitoring tool provides real-time visibility into the infrastructure you create, it allows you to instantly detect, troubleshoot, and resolve system errors/outliers or security threats before they impact the users. What it does is that it centralizes log data from diverse systems like CentOS, Ubuntu, and Docker, making it easy to audit for compliance, trace root causes, and automate alerts across the environment. This proactively minimizes downtime and streamlines performance management across your entire network.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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